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Data Analytics with R

Quality Laboratory Manual

Experiment No. 09

To implement and apply logistic regression using `glm()` in R to classify binary outcomes



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Experiment No. 09

Title of Experiment: To implement and apply logistic regression using `glm()` in R to classify binary outcomes

Aim of Experiment: To build and apply a logistic regression model in R using the `glm()` function for binary classification problems and interpret classification results

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Hard disk: 10 GB free space
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- R programming language (latest stable version)
- RStudio IDE
- Spreadsheet software (Microsoft Excel/ Libre Office)
- Built-in R graphics package
- Internet Connection (for dataset access)/ CSV dataset file

Experiment Objectives:

- To understand the concepts of binary classification and working of logistic regression
- To use `glm()` function with binomial family and predict probabilities and class labels
- To interpret logistic regression coefficients and evaluate classification outputs
- To apply logistic regression to real-world datasets

Experiment Outcomes:

After completing the experiment, students will be able to –

- Build logistic regression models using R
- Perform binary classification tasks and predict categorical outcomes
- Interpret probability-based predictions and understand decision boundaries
- Apply classification techniques in analytics applications

Theory:

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems where the output variable is categorical, especially binary (Yes/No, 0/1, Pass/Fail). Unlike linear regression, logistic regression predicts probabilities instead of continuous numeric values.

It is widely used in:

- Disease prediction
- Spam detection
- Customer churn prediction
- Fraud detection
- Student performance classification

Binary Classification

In binary classification, the dependent variable has only two possible outcomes:

Value	Meaning
0	Negative / No / Fail
1	Positive / Yes / Pass

Logistic regression estimates the probability that a data point belongs to class 1

Why Linear Regression is Not Suitable for Classification

Linear regression can produce values outside the range [0,1], which is not suitable for probabilities. Logistic regression overcomes this limitation using the Sigmoid Function, which maps outputs between 0 and 1

The sigmoid (logistic) function converts any real value into probability

$$P(Y = 1) = \frac{1}{1 + e^{-(a+bX)}}$$

Where;

$P(Y=1)$ = Probability of positive class

a = Intercept

b = Coefficient

X = Independent variable

The output always lies between 0 and 1

Logistic Regression using glm() in R

R provides the glm() (Generalized Linear Model) function for logistic regression

```
model <- glm(y ~ x, data = dataset, family = "binomial")
```

where, family = "binomial" specifies logistic regression

Probability Prediction

The model predicts probabilities using:

```
predict(model, type = "response")
```

Typical classification rule:

Probability > 0.5 → Class 1

Probability $\leq 0.5 \rightarrow$ Class 0

Interpretation of Coefficients

Positive coefficient $\rightarrow$ increases probability of class 1

Negative coefficient $\rightarrow$ decreases probability of class 1

The coefficients represent change in log-odds.

Applications of Logistic Regression

Logistic regression is widely used in:

- Medical diagnosis
- Banking fraud detection
- Loan approval systems
- Email spam filtering
- Marketing response prediction

Task to Perform

Predicting customer loan approval using logistic regression

A financial institution wants to automate its loan approval process. Based on previous customer records, the bank aims to predict whether a customer is likely to get loan approval depending on their income and credit score.

As a Data Analyst, you are assigned to:

- Build a Logistic Regression model
- Classify loan approval outcomes
- Predict probabilities of approval
- Interpret the model results

Dataset

```
loan_data = data.frame(
  Income = c(
    25, 30, 28, 35, 40, 45, 50, 55, 60, 65,
    70, 75, 80, 85, 90, 95, 100, 105, 110, 115
  ),

  CreditScore = c(
    550, 580, 600, 620, 640, 650, 670, 690, 700, 720,
    730, 740, 750, 760, 770, 780, 790, 800, 810, 820
  ),

  LoanApproved = c(
    0, 0, 0, 0, 0, 0, 1, 1, 1, 1,
    1, 1, 1, 1, 1, 1, 1, 1, 1, 1
  )
)
```

Task 1: Data Understanding

- Display dataset
- Identify:
 - Independent variables
 - Dependent variable

Task 2: Build Logistic Regression Model

Use:

- Income
- CreditScore

to predict:

- LoanApproved

using glm()

Task 3: Display Model Summary

Interpret:

- Coefficients
- Significance
- Relationship direction

Task 4: Predict Probabilities

Predict probabilities using predict() function

Task 5: Convert Probabilities into Classes

Use threshold:

- Probability > 0.5 → Approved (1)
- Else → Not Approved (0)

Task 6: Prediction for New Customer

Predict loan approval for:

Income	Credit Score
68	710

Task 7: Interpretation

Answer the following questions:

- 1) How does income affect approval?
- 2) How does credit score affect approval?
- 3) Which customers are more likely to get approval?

Conclusion:

The experiment successfully demonstrated the implementation of logistic regression using the `glm()` function in R. students learned how to build binary classification models, predict probabilities, interpret coefficients and classify outcomes. Logistic regression is a fundamental algorithm used extensively in predictive analytics and machine learning applications

Expected Oral Questions:

- 1) What is logistic regression?
- 2) Difference between linear and logistic regression?
- 3) Why use sigmoid function?
- 4) What is binary classification?
- 5) What does family="binomial" mean?
- 6) What is probability threshold?
- 7) Why can't linear regression be used for classification?

FAQs in Interview:

- 1) What is the main purpose of logistic regression?
- 2) Why is sigmoid function important?
- 3) What is decision boundary?
- 4) What evaluation metrics are used for logistic regression?
- 5) What is the difference between probability and class prediction?